BIRLA INSTITUTE OF TECHNOLOGY

Mesra, Ranchi

NEP-2020 CURRICULUM BOOK

(Effective from Academic Session: Monsoon 2024)

Bachelor of Technology

DEPARTMENT OF

ELECTRONICS AND COMMUNICATION ENGINEERING

INSTITUTE VISION

To become a Globally Recognized Academic Institution in consonance with the social, economic and ecological environment, striving continuously for excellence in education, research, and technological service to the National needs.

INSTITUTE MISSION

To educate students at Under Graduate, Post Graduate, Doctoral, and Post-Doctoral levels to perform challenging engineering and managerial jobs in industry.

To provide excellent research and development facilities to take up Ph.D. programmes and research projects.

To develop effective teaching learning skills and state of art research potential of the faculty.

To build national capabilities in technology, education, and research in emerging areas.

To provide excellent technological services to satisfy the requirements of the industry and overall academic needs of society.

DEPARTMENT VISION

To become a centre of excellence in teaching and research for creating technical manpower to meet the technological, societal and environmental needs of the country in the field of Electronics and Communication Engineering.

DEPARTMENT MISSION

DM1: To offer state of the art education of global standards through innovative methods of teaching and learning with practical orientation aiming to prepare the students for successful career and to provide required technological services.

DM2: To prepare the students to think independently, take initiative, lead a team in an organization, take responsibility and solve the problems related to industry, society, environmental, health, safety, legal and cultural issues maintaining the professional ethics.

DM3: To pursue high quality contemporary research through continued interaction with research organizations and industries.

PROGRAMME EDUCATIONAL OBJECTIVES (PEOs)

PEO1: To develop an ability to apply the knowledge acquired in basic sciences and engineering for solving Electronics and Communication Engineering problems with regards to technical, economic, environmental and social contexts.

PEO2: To build confident and competent graduates capable of designing and testing electronic devices and communication systems for the given specifications.

PEO3: To inculcate an attitude to work in a team using technical knowhow, software tools and interdisciplinary working environments to achieve project goals.

PEO4: To nurture effective communication and interpersonal skills to demonstrate leadership qualities and exhibit professional ethics.

PEO5: To develop an ability, enthusiasm and understanding for life-long learning development and motivation towards higher studies and research.

PROGRAMME OUTCOMES (POs)

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

Knowledge and Attitude Profile (WK)

WK1: A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.

WK2: Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.

WK3: A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.

WK4: Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.

WK5: Knowledge, including efficient resource use, environmental impacts, whole-life cost, re use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.

WK6: Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.

WK7: Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.

WK8: Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.

WK9: Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

PROGRAMME SPECIFIC OUTCOMES (PSOs)

PSO1: Apply the knowledge acquired in basic sciences and engineering for solving electronic and communication engineering problems.

PSO2: Build competence in design and analysis of electronics and communication systems.

PSO3: Develop skills to carry out research in electronic instrumentation, signal processing, VLSI systems, microwave engineering, wireless communication and networking.

Mapping of POs and PSOs with PEOs

	PEO1	PEO2	PEO3	PEO4	PEO5
PO1	3	3	2	2	2
PO2	3	3	2	2	2
PO3	3	3	2	2	2
PO4	3	3	2	2	2
PO5	3	3	2	2	2
PO6	2	2	3	3	3
PO7	3	2	3	3	3
PO8	2	3	3	3	1
PO9	2	3	3	3	2
PO10	1	2	2	1	2
PO11	3	2	1	1	3
PSO1	3	3	2	1	2
PSO2	3	3	2	1	2
PSO3	3	2	1	1	3

Grading: No correlation – 0, Low correlation - 1, Moderate correlation – 2, High Correlation - 3

Mapping of Department Mission (DM) with PEOs

	DM1	DM2	DM3
PEO1	3	2	3
PEO2	3	3	3
PEO3	3	3	2
PEO4	2	2	2
PEO5	2	2	3

Grading: No correlation – 0, Low correlation - 1, Moderate correlation – 2, High Correlation - 3

Course Structure for B.Tech. (Electronics and Communication Engineering)

Based on NEP-2020, CBCS and OBE, Effective from 2024-2025

FIRST YEAR

Sr. No.	Sem.	Cat.	Course Code / Subjects	L	T	P	C
I.1	I	FS	MA24101 - Mathematics - I	3	1	0	4
I.2	I		CH24101 - Chemistry	3	1	0	4
I.3	I	GE	EC24101 - Basic Electronics	2	1	0	3
I.4	I		ME24101 - Basics of Mechanical Engineering	2	1	0	3
I.5	I	FS	CE24101 - Environmental Science	2	0	0	2
I.6	I	FS	CH24102 - Chemistry Lab	0	0	2	1
I.7	I	GE	EC24102 - Basic Electronics Lab	0	0	2	1
I.8	I		ME24102 - Engineering Graphics	0	0	4	2
I.9	I		PE24102 - Workshop Practice	0	0	2	1
I.10	I	MC	MC24 101/102/103/104/105 - Choice of: NCC/NSS/PT & Games/Creative Arts (CA)/Entrepreneurship	0	0	2	1
	II	FS	MA24103 - Mathematics - II	3	1	0	4
	II		PH24101 - Physics	3	1	0	4
	II		BE24101 - Biological Science for Engineers	2	0	0	2
	II	GE	CS24101 - Programming for Problem Solving	3	1	0	4
	II		EE24101 - Basics of Electrical Engineering	2	1	0	3
	II	FS	PH24102 - Physics Lab	0	0	2	1
	II	GE	CS24102 - Programming for problem Solving Lab.	0	0	2	1
	II		EE24102 - Electrical Engineering Lab.	0	0	2	1
	II	HSS	HS24131 - Communication Skill - I	0	0	3	1.5
	II	MC	MC24 106/107/108/109/110 - Choice of: NCC/NSS/PT & Games/Creative Arts (CA)/Entrepreneurship	0	0	2	1

GRAND TOTAL FOR FIRST YEAR: 44.5 Credits

SECOND YEAR

Sr. No.	Sem.	Cat.	Course Code / Subjects	L	T	P	C
III.1	III	PC	EC24201 - Electronic Devices	3	1	0	4
III.2	III		EC24203 - Digital System Design	3	0	0	3
III.3	III		EC24205 - Network Theory	3	1	0	4
III.4	III		EC24207 - Signals and Systems	3	0	0	3
III.5	III		EC24209 - Probability and Random Processes	3	0	0	3
III.6	III	HSS	MT24131 - UHV-II: Understanding Harmony	3	0	0	3
III.7	III	PC	EC24202 - Electronic Devices Lab	0	0	2	1
III.8	III	PC	EC24204 - Digital System Design Lab	0	0	2	1
III.9	III	MC	MC24 201/202/203/204/205 - Choice of: NCC/NSS/PT & Games/Creative Arts (CA)/Entrepreneurship	0	0	2	1
IV.1	IV	PC	EC24251 - Analog Circuits	3	1	0	4
IV.2	IV		EC24253 - Analog Communication	3	1	0	4
IV.3	IV		EC24255 - Computer Architecture	3	0	0	3
IV.4	IV		EC24257 - VLSI Design	3	1	0	4
IV.5	IV	PE	XX24XXX/MO24201 - Open Elective - I/MOOC - I	3	0	0	3

IV.6	IV	HS	HS24211 - Indian Knowledge System	2	0	0	0
IV.7	IV	PC	EC24252 - Analog Circuits Lab	0	0	2	1
IV.8	IV	PC	EC24258 - VLSI Design Lab	0	0	2	1
IV.9	IV	MC	MC24 206/207/208/209/210 - Choice of: NCC/NSS/PT & Games/Creative Arts (CA)/Entrepreneurship	0	0	2	1

GRAND TOTAL FOR SECOND YEAR: 44 Credits

THIRD YEAR

Sr. No.	Sem.	Cat.	Course Code / Subjects	L	T	P	C
V.1	V	PC	EC24301 - Electromagnetic Fields and Waves	3	1	0	4
V.2	V	PC	EC24303 - Digital Communication	3	0	0	3
V.3	V	PC	EC24305 - Microprocessors	3	0	0	3
V.4	V	PC	EC24307 - Data Communication and Computer Networking	3	1	0	4
V.5	V	PE	EC24XXX - Program Elective-I (PE-I)	3	0	0	3
V.6	V	OE	XX24XXX/MO24301 - Open Elective - II / MOOC - II	3	0	0	3
V.7	V	PC	EC24304 - Communication System Lab	0	0	2	1
V.8	V	PC	EC24306 - Microprocessors Lab	0	0	2	1
V.9	V	PC	EC24300 - Project - I				2
VI.1	VI	PC	EC24351 - Digital Signal Processing	3	1	0	4
VI.2	VI	PC	EC24353 - Control Systems	3	1	0	4
VI.3	VI	PC	EC24355 - Embedded Systems	3	0	0	3
VI.4	VI	PE	EC24XXX - Program Elective-II (PE-II)	3	0	0	3
VI.5	VI	OE	XX24XXX/MO24303 - Open Elective - III / MOOC - III	3	0	0	3
VI.6	VI	HSS	MT24204 - Constitution of India	2	0	0	0
VI.7	VI	PC	EC24352 - Digital Signal Processing Lab	0	0	2	1
VI.8	VI	PC	EC24356 - Embedded Systems Lab	0	0	2	1
VI.9	VI	PE	EC243XXX - Program Elective-II Lab	0	0	2	1
VI.10	VI	PC	EC24350 - Project - II				2
VI.11	VI	HSS	HS24133 - Communication Skill - II	0	0	3	1.5

GRAND TOTAL FOR THIRD YEAR: 47.5 Credits

FOURTH YEAR

Sr. No.	Sem.	Cat.	Course Code / Subjects	L	T	P	C
VII.1	VII	PC	EC24401 - Microwave Theory and Techniques	3	1	0	4
VII.2	VII	PE	EC24XXX - Program Elective-III (PE-III)	3	0	0	3
VII.3	VII	PE	EC24XXX - Program Elective-IV (PE-IV)	3	0	0	3
VII.4	VII	PE	EC24XXX - Program Elective-V (PE-V)	3	0	0	3
VII.5	VII	OE	XX24XXX/MO24401 - Open Elective - IV / MOOC - IV	3	0	0	3
VII.6	VII	PC	EC24402 - Microwave Lab	0	0	2	1
VII.7	VII	MC	MC24400 - Summer Training (Minimum Four Weeks / 160 Hrs)				4
VII.8	VII	PC	EC24400 - Project - III				3

VIII.1	VIII	PC	EC24450/EC24490 - Project-IV / Industry Internship				6
VIII.2	VIII		EC24498 - Comprehensive Viva				2

GRAND TOTAL FOR FOURTH YEAR: 32 Credits

GRAND TOTAL FOR B.TECH.: 168 Credits

Vocational Courses

Exit after 1st Year:

Course Name	L	T	P	C
EC24190 - Analog and Digital Circuit Design	1	0	4	3
EC24192 - Electronic Equipments Servicing and Maintenance	1	0	4	3

Exit after 2nd Year:

Course Name	L	T	P	C
EC24290 - Electronic Instruments and Measurements	1	0	4	3
EC24292 - Communication Engineering	1	0	4	3

List of Programme Elective (PE) Courses

[ECE students to choose one from each group]

Sem.	Code	Course Name	L	T	P	C
V	EC24309	Information Theory and Coding	3	0	0	3
V	EC24311	ASIC Design	3	0	0	3
V	EC24313	Artificial Intelligence and Machine Learning	3	0	0	3
VI	EC24357	Fiber Optic Communication	3	0	0	3
VI	EC24359	Wireless Communication	3	0	0	3
VI	EC24361	Electronic Measurements	3	0	0	3
VI	EC24358	Fiber Optic Communication Lab	0	0	2	1
VI	EC24360	Wireless Communication Lab	0	0	2	1
VI	EC24362	Measurement and Instrumentation Lab	0	0	2	1
VII	EC24403	Mobile & Cellular Communication	3	0	0	3
VII	EC24405	Satellite Communication	3	0	0	3
VII	EC24407	Industrial Instrumentation	3	0	0	3
VII	EC24409	Optimization Techniques	3	0	0	3
VII	EC24411	Internet of Things	3	0	0	3
VII	EC24413	Low Power VLSI Circuits	3	0	0	3
VII	EC24415	Speech and Audio Processing	3	0	0	3

VII	EC24417	FPGA System	3	0	0	3
VII	EC24419	Antennas and Wave Propagation	3	0	0	3
VII	EC24421	Microelectronic Devices and Circuits	3	0	0	3
VII	EC24423	Nanoelectronics	3	0	0	3
VII	EC24425	MEMS and Microsystems	3	0	0	3
VII	EC24427	Radar Engineering	3	0	0	3
VII	EC24429	Wireless Sensors Network	3	0	0	3
VII	EC24431	Digital Image Processing	3	0	0	3
VII	EC24433	Bio-Medical Electronics & Signal Processing	3	0	0	3
VII	EC24435	Deep Learning	3	0	0	3
VII	EC24437	Electronic Packaging	3	0	0	3

List of Open Elective (OE) Courses

[Open Electives to be opted only by other departments' students]

Sem.	Code	Course Name	Prerequisites	L	T	P	C
IV	EC24259	Sensors and Transducers	N/A	3	0	0	3
IV	EC24261	Digital Computer Electronics	N/A	3	0	0	3
IV	EC24263	Introduction to VLSI Design	N/A	3	0	0	3
V	EC24321	Introduction to Communication System	N/A	3	0	0	3
V	EC24333	Introduction to Microcontrollers	N/A	3	0	0	3
VI	EC24363	Introduction to MEMS	N/A	3	0	0	3
VI	EC24379	Measurement & Instrumentation	N/A	3	0	0	3
VII	EC24439	Introduction to Signal Processing	N/A	3	0	0	3
VII	EC24441	Electronics Technology	N/A	3	0	0	3

NEP-2020 Guidelines for Multiple Exit and Multiple Entry

Multiple Exit

Year of Exit	Certification	Credits Required
First Year	Certificate in Engineering	Must have passed 1st year + *6 credits for vocational courses of level-1
Second Year	Diploma in Engineering (Electronics & Communication Engineering)	Must have passed 1st and 2nd year + *6 credits for vocational courses of level-2
Third Year	B Sc. Engineering (Electronics & Communication Engineering)	Must have passed 1st, 2nd and 3rd year

Multiple Entry

Will be based on admission test against vacant seats, considering the past academic records/ABC/CGPA/JEE(Main)/Closing Rank for the corresponding admission year/NIRF/Institute Reputation as well as without violating Institute own admission policy.

Course Structure for In-depth Specialization for 2024 Batch and onwards

[To be provided]

Course Structure for Minor in ECE for 2024 Batch and onwards

[To be provided]